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Analysis PSet #7

1. Converge.

By Ruler 3.29,

$\sum_{n\geq 2} \frac{1}{n(\ln(n))^2}$ converges.

$\Rightarrow \sum_{n\geq 3} \frac{1}{n(\ln(n))^2}$ converges.

For $n \geq 3$, $\ln(n) < \log_2(n)$ and $\ln(n), \log_2(n) > 1$

$\Rightarrow \frac{1}{n(\log_2(n))^2} < \frac{1}{n\ln(n)^2}$

By comp. test for conv. $\sum_{n\geq 3} \frac{1}{n(\log_2(n))^2}$ converges.

$\because \sum_{n\geq 2} \frac{1}{n(\log_2(n))^2} = \frac{1}{2} + \sum_{n\geq 3} \frac{1}{n(\log_2(n))^2}$ where $\frac{1}{2}$ is finite

$\therefore \sum_{n\geq 2} \frac{1}{n(\log_2(n))^2}$ converges.

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2. a) $\sum \frac{(n+1)^3 z^{n+1}}{n^3 z^n} = \sum z^n \frac{(n+1)^3}{n^3}$

$\Rightarrow R = 1$

b) $\sum \frac{2^{n+1} \cdot z^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n \cdot z^n} = \sum 2^n z^n \cdot \frac{1}{(n+1)}$

$R = \frac{1}{2}$

2.

a) $\lim_{n\to\infty} \left(n^3\right)^{\frac{1}{n+1}} = \lim_{n\to\infty} (1)^3 = 1$

$R = 1$

b) $\lim_{n\to\infty} \left(\frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n}\right) = \lim_{n\to\infty} \frac{2}{n+1} = 0$

$R = \infty$

c) $\lim_{n\to\infty} \left(\left(\frac{2^n}{n^2}\right)^{\frac{1}{n}}\right) = \lim_{n\to\infty} \frac{2}{1^2} = 2$

$R = \frac{1}{2}$

d) $\lim_{n\to\infty} \left(\left(\frac{n^3}{3^n}\right)^{\frac{1}{n}}\right) = \frac{1}{3}$ $R = 3$

3.

a) Yes, the sequence looks something like $(1-1)+(2-2)+(3-3)$ in the partial sums in this property. Then, the seq. of partial sums is (x_n); $x_n = 0$. which is bounded.

Proof outline,

Evidently, A converges to 0 (b/c $\exists$ always $a_n - m$ for each m term, a_{n+1})!

b) Now, fixing $n=k$, each term $\frac{k-(k-1)}{k} \geq \frac{1}{k}$ for $k \geq 2$ and $\frac{1}{k}$ for $n \geq 1$.

The seq. B then $\left(\sum_{n\geq 1} \frac{1}{n}\right)$ which diverges,

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$\forall \varepsilon > 0$, $\exists N$ s.t. ε for $n \geq N$,

$\left|\left(\frac{x_n}{y_n} - L\right)\right| < \varepsilon$. $x_n, y_n > 0 \Rightarrow x_n > y_n$

Let $\exists$ choose $\varepsilon = L$ and the associated N.

$\Rightarrow \left|\frac{x_n}{y_n}\right| < 2L \Rightarrow -2Ly_n < x_n < 2Ly_n$

$\sum_{n\geq 1} 2Ly_n = 2L\left(\sum_{n\geq 1} y_n\right)$ which converges b/c L is finite.

$x_n < 2Ly_n$ & by comparison test, $\sum_{n\geq 1} x_n$ converges.

~~Only If~~

$-2L y_n < x_n \Rightarrow y_n < -\frac{1}{2L} x_n$

$\sum_{n=1}^{\infty} -\frac{1}{2L} x_n = -\frac{1}{2L} \sum_{n=1}^{\infty} x_n$ and

$-\frac{1}{2L}$ is finite, so $-\frac{1}{2L} \sum_{n=1}^{\infty} x_n$ converges.

$y_n < -\frac{1}{2L} x_n \Rightarrow \sum_{n=1}^{\infty} y_n$ converges.